

Notes: Goyal & Saretto (RFS, 2025)
Can Equity Option Returns Be Explained by a Factor Model? IPCA Says Yes

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1 Big picture

- **Question.** Do the large average returns documented for delta-hedged equity option strategies represent true abnormal profits, or can they be explained by an option-specific factor model?
- **Main answer.** A three-factor IPCA model explains most of the profitability of many previously documented option strategies.
- **Setting.** The empirical object is expiration-to-expiration, daily delta-hedged call option returns from 1996 to 2022.
- **Predictors.** The paper uses 46 characteristics from prior option-return and stock-return literature.
- **Raw return puzzle.** The RV–IV strategy earns 2.87% per month before transaction costs and 1.73% net of transaction costs.
- **Main pricing result.** The average IPCA alpha across 46 long-short strategies is about 6 basis points per month.
- **Economic takeaway.** Evidence on option return predictability does not necessarily imply option-market inefficiency; it may reflect exposure to systematic option-return factors.

2 Data and measurement

The sample covers individual equity options from 1996 to 2022. Returns are expiration-to-expiration returns: positions are initiated on the Monday after expiration Friday and held until the next expiration Friday. Call positions are delta-hedged daily.

The predictor library contains 46 characteristics: contract characteristics, risk-neutral distribution variables, physical distribution variables, and firm characteristics. Long-short strategies sort stocks into deciles and use the extreme deciles. The direction is chosen so the full-sample return is positive.

Interpretation. Transaction costs are especially important in options. The paper adjusts initial option prices using an effective-to-quoted spread ratio of 30% and evaluates net returns separately.

3 Models and empirical specifications

3.1 IPCA return equation

$$R_{i,t+1} = \alpha_{i,t} + \beta'_{i,t} F_{t+1} + \varepsilon_{i,t+1} = Z'_{i,t} \Gamma_{\alpha} + Z'_{i,t} \Gamma_{\beta} F_{t+1} + \varepsilon_{i,t+1}.$$

Interpretation. IPCA makes both alpha and factor loadings conditional on observable characteristics.

3.2 Matrix form

$$R_{t+1} = Z_t \Gamma_{\alpha} + Z_t \Gamma_{\beta} F_{t+1} + \varepsilon_{t+1}, \quad W_t = Z'_t Z_t / N_{t+1}, \quad X_{t+1} = Z'_t R_{t+1} / N_{t+1}.$$

Interpretation. The model can be estimated with low-dimensional managed-portfolio returns rather than the full option panel.

3.3 Restricted model, alpha, and pricing metrics

The restricted model imposes $\Gamma_\alpha = 0$. Strategy alpha is

$$\hat{\alpha} = \frac{1}{T} \sum_t \left(R_{t+1}^{portfolio} - Z_t' \hat{\Gamma}_\beta \hat{F}_{t+1} \right).$$

Pricing metrics include total R^2 , time-series R^2 , cross-section R^2 , and relative pricing error.

3.4 Characteristic importance

$$Impact_l = \sqrt{\sum_{k=1}^K \gamma_{\beta,l,k}^2}.$$

Interpretation. The statistic measures how strongly a characteristic determines factor exposures across all factors.

4 Identification and empirical design

This is an asset-pricing benchmark exercise, not a causal design. The credibility of the argument comes from out-of-sample fit, multiple-testing adjustment, transaction-cost evaluation, and restricted-characteristic tests in which target characteristics are excluded from IPCA estimation.

5 Tables and figures

All visuals are placed only in this section. Each image is directly cropped from the paper.

5.1 Table 1: Long-short delta-hedged call portfolios

Read: Table 1 establishes the anomaly set: many characteristic-sorted option strategies have large raw returns, and 16 remain significant after transaction costs. RV-IV earns 2.87% gross and 1.73% net per month.

Interpretation. This table is the benchmark that IPCA must explain. It is evidence of strong predictability, but not yet evidence of mispricing because the returns have not been adjusted by an option-specific factor model.

Economic message. The raw evidence makes option strategies look highly profitable; the rest of the paper asks whether those profits are systematic risk compensation.

5.2 Table 2: IPCA performance

Read: The three-factor model delivers managed-portfolio total R^2 of 87.60% in-sample and 83.27% out-of-sample. Managed-portfolio cross-section R^2 falls only moderately from 75.70% to 69.62%.

Interpretation. Table 2 validates the model as a pricing benchmark. High managed-portfolio fit matters because the long-short strategies are characteristic-managed objects.

Table 1
Long-short delta-hedged call portfolios

	Construction	Return		Return w/ TC	
Moneynews	1-10	0.38	(3.60)	-0.86	(-8.13)
Bid-ask spread	1-10	0.71	(8.07)	-0.13	(-1.50)
Open interest	1-10	0.04	(0.66)	-0.59	(-9.84)
Delta	10-1	0.22	(2.58)	-0.68	(-7.52)
Vega	10-1	1.44	(14.83)	0.47	(5.02)
Gamma	1-10	0.67	(7.09)	-0.25	(-2.69)
Volume	10-1	0.23	(3.11)	-0.40	(-5.28)
Option price	10-1	0.65	(5.72)	-0.12	(-1.09)
IV ATM	1-10	2.34	(17.94)	1.40	(10.95)
IV slope	10-1	2.15	(25.53)	0.91	(10.46)
IV term	10-1	1.42	(15.94)	0.43	(4.84)
IV vol	1-10	1.45	(15.65)	0.61	(6.81)
MFvol	1-10	1.16	(5.16)	0.69	(3.13)
MFskew	1-10	0.83	(4.98)	0.42	(2.54)
MFkurt	10-1	0.74	(4.41)	0.33	(1.97)
Stock return	10-1	0.04	(0.46)	-0.93	(-9.30)
Stock return11	10-1	0.29	(2.91)	-0.71	(-7.09)
RV	1-10	1.06	(9.23)	0.22	(1.94)
Rskew	1-10	0.14	(2.23)	-0.74	(-12.62)
Rkurt	1-10	0.91	(14.48)	0.15	(2.45)
Turnover	1-10	0.13	(1.32)	-0.66	(-6.68)
IdiosyncVol	1-10	1.15	(11.18)	0.33	(3.26)
Max10	1-10	0.94	(8.35)	0.11	(0.99)
Autocorrelation	10-1	0.07	(1.24)	-0.72	(-12.25)
RV-IV	10-1	2.87	(24.51)	1.73	(14.94)
RV-MFvol	10-1	1.64	(7.49)	1.11	(5.16)
Rskew-MFskew	1-10	0.11	(0.73)	-0.31	(-2.13)
Rkurt-MFkurt	1-10	1.00	(8.03)	0.65	(5.22)
BM	10-1	0.13	(1.39)	-0.75	(-8.33)
Profitability	10-1	0.92	(11.71)	0.03	(0.41)
InstOwn	10-1	0.93	(12.10)	0.03	(0.38)
MarketCap	10-1	1.59	(16.74)	0.67	(7.22)
RSI	10-1	0.10	(1.56)	-0.63	(-9.77)
Assets	10-1	1.46	(14.59)	0.64	(6.71)
Debt	10-1	0.68	(8.82)	0.04	(0.50)
Leverage	10-1	0.00	(0.06)	-0.84	(-15.11)
CashFlowVar	1-10	0.72	(9.11)	-0.01	(-0.16)
Cash to asset	1-10	0.87	(9.42)	-0.04	(-0.43)
AnalystDisp	10-1	0.52	(8.18)	-0.49	(-7.82)
1yr NewIss	1-10	0.43	(5.58)	-0.34	(-4.45)
5yr NewIss	1-10	0.65	(8.33)	-0.09	(-1.25)
Profit margin	10-1	1.15	(13.37)	0.25	(3.05)
Stock price	10-1	1.73	(16.09)	0.74	(7.03)
ROE	10-1	0.82	(11.09)	-0.07	(-0.99)
ExternalFin	1-10	0.52	(6.10)	-0.31	(-3.70)
Z-score	10-1	0.68	(7.99)	-0.29	(-3.62)

The table reports average returns (in percent per month) with and without accounting for transaction costs of 46 long-short portfolios. We construct the trading strategies so that average returns are positive by either buying options positions in decile 10 and writing option positions in decile 1, or by buying options in decile 1 and writing options in decile 10. We tabulate the portfolio construction in the second column. Transaction costs are calculated using a ratio of effective to quoted spread equal to 30%. t -statistics larger than the [Benjamini and Hochberg \(1995\)](#) 5% FDR adjusted threshold (i.e., 2.25 for mean return and 2.44 for transaction cost adjusted returns) are reported in bold. The sample period is 1996 to 2022.

Table 2
IPCA performance

	Number of factors				
	1	2	3	4	5
	A. In-sample				
Stock-level option positions:					
Total R^2 (%)	8.77	11.54	12.22	12.75	13.22
Time-series R^2 (%)	12.11	15.04	15.68	15.87	16.16
Cross-section R^2 (%)	8.69	10.84	11.38	11.86	12.27
Relative pricing error	78.46	73.68	72.74	71.80	71.20
Managed portfolios:					
Total R^2 (%)	63.93	82.45	87.60	89.41	91.23
Time-series R^2 (%)	40.65	65.15	73.43	76.48	80.33
Cross-section R^2 (%)	53.45	70.50	75.70	78.38	80.58
Relative pricing error	24.46	0.38	0.23	0.18	0.15
Bootstrap test:					
W_0 p-value	.00	.01	.00	.00	.03
Economic test:					
# Unpriced factors	0	0	0	1	3
	B. Out-of-sample				
Stock-level option positions:					
Total R^2 (%)	6.70	8.74	9.32	9.77	10.26
Time-series R^2 (%)	0.08	-6.76	-6.15	-9.12	-19.25
Cross-section R^2 (%)	6.65	8.32	8.87	9.28	9.77
Relative pricing error	81.75	77.73	76.99	76.39	75.32
Managed portfolios:					
Total R^2 (%)	54.39	78.23	83.27	85.01	87.65
Time-series R^2 (%)	28.22	57.52	66.42	68.95	73.94
Cross-section R^2 (%)	47.19	63.92	69.62	72.14	75.00
Relative pricing error	48.16	9.47	5.26	6.06	5.16

The table presents performance measures of IPCA models for delta-hedged call returns. We report total, time-series, and cross-section R^2 , as well as the relative pricing error (all in percent) from Equation (8) for individual stock option positions and managed portfolios for the constrained ($\Gamma_{\pi} = 0$) model. We also report p -values, in decimal units, on the bootstrap Wald test of $\Gamma_{\pi} = 0$. The last row of the top panel reports number of unpriced factors calculated as the number of estimated IPCA factors whose returns are statistically indistinguishable from zero. In panel B, we report out-of-sample performance measures, obtained by splitting the sample in half and using the first half to estimate the model. We then use estimated factors and conditional betas to produce a 1-month out-of-sample forecast. We then roll the estimation procedure forward 1 month at a time, until the end of the sample. Results are reported for IPCA models with $K = 1$ to 5. Conditional betas are estimated for each of the characteristics used in Table 1. The sample period is 1996 to 2022.

Economic message. A parsimonious three-factor IPCA model captures much of systematic variation in option returns.

5.3 Table 3 and Figure 1: IPCA alphas

Table 3
IPCA alphas

Moneyness	-0.05	(-0.52)	Autocorrelation	0.08	(1.34)
Bid-ask spread	-0.01	(-0.08)	RV-IV	0.34	(4.68)
Open interest	0.06	(0.86)	RV-MFvol	-0.27	(-1.24)
Delta	0.09	(0.90)	Rskew-MFskew	-0.04	(-0.24)
Vega	0.08	(0.58)	Rkurt-MFkurt	0.21	(1.51)
Gamma	0.04	(0.28)	BM	0.15	(2.10)
Volume	0.05	(0.65)	Profitability	0.02	(0.25)
Option price	-0.19	(-1.14)	InstOwn	0.03	(0.46)
IV ATM	0.24	(2.11)	MarketCap	0.11	(1.00)
IV slope	0.34	(4.58)	RSI	0.12	(1.71)
IV term	0.16	(2.11)	Assets	0.14	(1.41)
IV vol	0.11	(1.21)	Debt	-0.03	(-0.36)
MFvol	-0.26	(-1.14)	Leverage	-0.01	(-0.15)
MFskew	0.11	(0.63)	CashFlowVar	0.02	(0.18)
MFkurt	-0.15	(-0.82)	Cash to asset	0.08	(1.14)
Stock return	0.13	(1.43)	AnalystDisp	-0.02	(-0.34)
Stock return11	0.03	(0.31)	1yr NewIss	0.09	(1.37)
RV	0.16	(1.41)	5yr NewIss	0.09	(1.34)
Rskew	-0.11	(-1.88)	Profit margin	0.04	(0.53)
Rkurt	0.14	(1.95)	Stock price	0.09	(0.65)
Turnover	0.06	(0.70)	ROE	0.06	(0.79)
IdiosynVol	0.16	(1.65)	ExternalFin	0.04	(0.51)
Max10	0.13	(1.24)	Z-score	0.05	(0.57)

The table presents IPCA alphas (in percent per month) and *t*-statistics in parentheses for long-short portfolios of delta-hedged call returns as described in Table 1. The IPCA model has 3 factors. Portfolios with *t*-statistics larger than the Benjamini and Hochberg (1995) 5% FDR adjusted threshold (i.e., 2.60 and 3.18 for mean return and alphas of long-short portfolios, respectively) are reported in boldface. The sample period is 1996 to 2022.

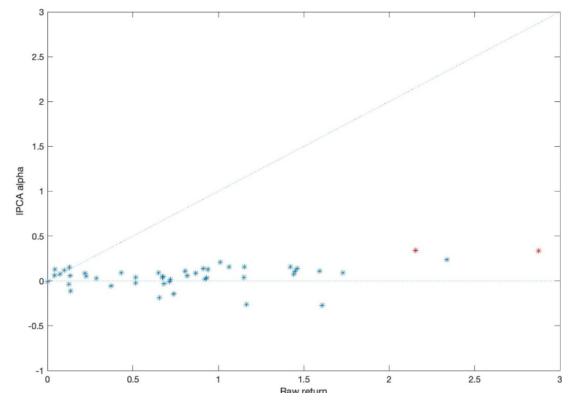


Figure 1
IPCA alphas

The figure plots portfolio alphas (in percent per month) from Table 3 against portfolio returns from Table 1 for long-short portfolios of delta-hedged call returns. The alpha is computed from the IPCA model with 3 factors. The sample period is 1996 to 2022.

ALT TEXT: Relationship between IPCA alphas and raw returns illustrates that IPCA alphas are close to zero for almost all strategies.

Read: The average alpha across 46 strategies is about 6 basis points. Only IV slope and RV-IV remain significant after FDR adjustment. Figure 1 shows most strategies near zero alpha.

Interpretation. IPCA compresses large raw returns into small residual alphas, showing that most profitability is explained by factor exposure.

Economic message. Most option strategy returns look less anomalous under an option-specific conditional factor model.

5.4 Figure 2 and Table 4: What drives IPCA betas?

Read: Figure 2 shows that RV-IV, Assets, and MarketCap are important gamma coefficients. Table 4 decomposes RV-IV decile returns and shows that Factor 1 explains most of the expected-return spread.

Interpretation. These visuals explain how a latent factor model becomes economically interpretable: characteristics determine betas, and betas determine expected returns.

Economic message. RV-IV is both a strong standalone signal and a core proxy for option-factor exposure.

5.5 Table 5 and Figure 3: Which characteristics matter?

Read: Table 5 ranks characteristics by beta-mapping impact; RV-IV, MarketCap, and Assets are central. Figure 3 shows that restricted IPCA models still price omitted-characteristic strategies well.

Interpretation. This guards against the critique that IPCA merely prices a strategy with its own signal.

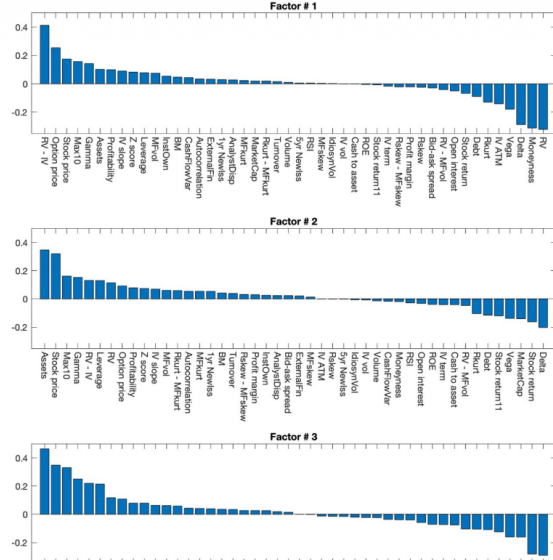


Figure 2
IPCA gammas
The figure shows IPCA gammas for each of the 3 factors in the IPCA model from Table 2. The sample period is 1996 to 2022.
ALT TEXT: Characteristic importance for each IPCA factor shows a large role for RV–IV, Assets, and MarketCap,

Table 5
Characteristics importance

RV–IV	0.54 (.00)	Open interest	0.12 (.02)
MarketCap	0.49 (.00)	Z-score	0.12 (.04)
Assets	0.46 (.00)	Idiosync Vol	0.11 (.07)
RV	0.45 (.05)	1yr Newlss	0.09 (.02)
Max 10	0.42 (.00)	MFskew	0.08 (.55)
Delta	0.42 (.03)	RSI	0.08 (.15)
Moneyness	0.40 (.02)	BM	0.08 (.17)
Stock price	0.37 (.08)	MFkurt	0.07 (.60)
Option price	0.29 (.69)	Autocorrelation	0.07 (.02)
Gamma	0.26 (.25)	Rkurt–MFkurt	0.07 (.45)
Vega	0.23 (.39)	CashFlowVar	0.06 (.08)
Turnover	0.22 (.01)	InstOwn	0.06 (.14)
Stock return	0.21 (.04)	ExternalFin	0.06 (.11)
Rkurt	0.18 (.00)	Profit margin	0.05 (.20)
RV–MFvol	0.17 (.36)	IV term	0.05 (.22)
Debt	0.16 (.06)	Bid-ask spread	0.05 (.52)
MFvol	0.16 (.48)	Rskew	0.05 (.16)
Leverage	0.16 (.07)	ROE	0.05 (.20)
IV ATM	0.15 (.45)	AnalystDisp	0.04 (.17)
Stock return11	0.14 (.15)	Rskew–MFskew	0.04 (.69)
IV slope	0.13 (.03)	Volume	0.03 (.06)
Profitability	0.13 (.04)	5yr Newlss	0.02 (.62)
Cash to asset	0.12 (.01)	IV vol	0.02 (.79)

The table presents measures of importance for each characteristic used in the IPCA estimation of Table 2 with 3 factors. Variables are sorted by the square root of the sum of squared Γ . We also report the p -value of the bootstrap test for each $\Gamma = 0$ in parentheses. The sample period is 1996 to 2022.

Table 4
RV–IV expected IPCA returns decomposition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ret	−2.89	−1.14	−0.81	−0.59	−0.52	−0.37	−0.25	−0.20	−0.06	−0.02
E(Ret)	−2.41	−1.46	−1.04	−0.75	−0.59	−0.36	−0.22	−0.11	0.02	0.13
α_{IPCA}	−0.48	0.32	0.24	0.16	0.07	−0.01	−0.03	−0.09	−0.07	−0.14
β_{F1} E(F1)	−2.63	−1.70	−1.27	−0.99	−0.84	−0.61	−0.49	−0.40	−0.31	−0.27
β_{F2} E(F2)	0.20	0.20	0.19	0.19	0.19	0.19	0.20	0.20	0.22	0.25
β_{F3} E(F3)	0.02	0.04	0.04	0.04	0.06	0.06	0.07	0.09	0.11	0.15
RV–IV	−1.75	−1.04	−0.68	−0.39	−0.11	0.13	0.39	0.68	1.04	1.75

The table presents a decomposition of the returns (in percent per month) of decile portfolios constructed by sorting stocks based on the difference between realized and implied volatility (RV–IV). Expected return and IPCA alpha are derived from the IPCA model estimated with 3 factors from Table 3. The sample period is 1996 to 2022.

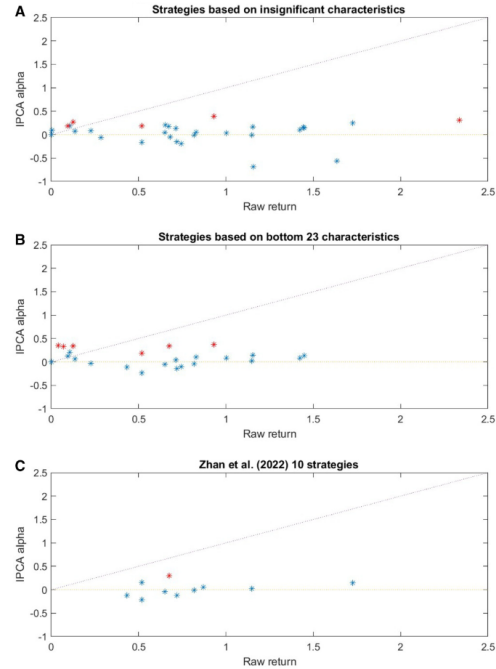


Figure 3
Characteristics subsamples

The figure shows IPCA alphas versus raw returns (both in percent per month) for three different set of characteristics. In each case, we retain some characteristics and estimate a restricted IPCA model which we use to price the long-short portfolios constructed with the characteristics that are not included in the estimation (i.e., target long-short portfolios). In panel A we use the 17 statistically significant characteristics from Table 5 to estimate the model and the remaining 29 statistically insignificant characteristics to construct the target long-short portfolios. In panel B we use the 23 characteristics with the highest impact score from Table 5 to estimate the model and the remaining 23 characteristics to construct the target long-short portfolios. In panel C, we use the 10 characteristics from Zhan et al. (2022) to construct the target portfolios and the remaining 36 to estimate the IPCA model. The sample period is 1996 to 2022.

ALT TEXT: IPCA alphas are close to zero for almost all strategies for different ways of estimating IPCA models (by retaining a subset of characteristics).

Economic message. The option-return cross-section is lower-dimensional than the 46-characteristic library suggests.

5.6 Table 6: Zhan et al. strategies

Read: Table 6 excludes the ten target characteristics from estimation and then prices strategies built from them. Raw returns remain strong, but restricted IPCA alphas are small and not significant after multiple-testing adjustment.

Interpretation. This is a stringent validation exercise because the model learns factor structure from other characteristics.

Economic message. Even newer option predictors are largely priced by the broader IPCA factor structure.

Table 6

Pricing the 10 portfolios in Zhan et al. (2022)

	Full sample 1996-2022		In-sample 1996-2016/5		Out-of-sample 2016/6-2022	
	Raw returns					
Stock price	1.73	(16.06)	1.55	(14.14)	2.27	(8.30)
CashFlowVar	0.72	(9.10)	0.50	(6.19)	1.41	(7.42)
Cash to asset	0.87	(9.41)	0.60	(6.13)	1.69	(8.47)
AnalystDisp	0.52	(8.17)	0.50	(6.68)	0.59	(4.80)
1yr NewIss	0.43	(5.57)	0.17	(2.14)	1.25	(7.10)
5yr NewIss	0.65	(8.31)	0.36	(5.10)	1.55	(7.54)
Profit margin	1.15	(13.35)	0.95	(11.17)	1.77	(7.94)
ROE	0.82	(11.07)	0.69	(9.04)	1.20	(6.58)
ExternalFin	0.52	(6.09)	0.26	(2.78)	1.31	(8.07)
Z-score	0.68	(7.98)	0.49	(5.72)	1.24	(5.86)
	Restricted IPCA alphas					
Stock price	0.15	(1.11)	0.05	(0.39)	0.43	(1.33)
CashFlowVar	-0.12	(-1.40)	-0.26	(-3.23)	0.31	(1.32)
Cash to asset	0.05	(0.68)	-0.03	(-0.33)	0.31	(1.80)
AnalystDisp	0.15	(2.61)	0.16	(2.36)	0.13	(1.12)
1yr NewIss	-0.13	(-1.97)	-0.22	(-3.43)	0.17	(1.06)
5yr NewIss	-0.04	(-0.69)	-0.16	(-2.53)	0.32	(1.95)
Profit margin	0.02	(0.30)	-0.04	(-0.46)	0.21	(1.26)
ROE	-0.01	(-0.18)	-0.08	(-1.08)	0.20	(1.27)
ExternalFin	-0.22	(-2.97)	-0.30	(-3.75)	0.05	(0.35)
Z-score	0.30	(3.13)	0.18	(1.77)	0.69	(2.96)

We exclude the 10 characteristics in Zhan et al. (2022) from the set of covariates and reestimate the IPCA model on the remaining 36 (plus a constant). We refer to this estimated model as the “restricted IPCA.” We then compute average raw returns and alphas (both in percent per month) in three sample periods: the full sample (January 1996 to December 2022), in-sample (January 1996 to April 2016) corresponding to the sample studied in Zhan et al. (2022), and out-of-sample (from May 2016 to December 2020). *t*-statistics larger than the Benjamini and Hochberg (1995) 5% FDR adjusted threshold are reported in bold. MHT thresholds for raw returns are 2.22, 2.06, and 3.11, for each of the three periods, respectively. MHT thresholds for restricted IPCA alphas are 3.12, 3.21, and 3.29, for each of the three periods, respectively.

Figure 1: Pricing the 10 portfolios in Zhan et al. (2022)

5.7 Figure 4 and Table 7: Factor interpretation

Read: Figure 4 maps factors to managed portfolios. Table 7 relates factor returns to stress states including market return, VIX, tail risk, volatility risk, skew risk, kurtosis risk, intermediary capital, average gamma, and average vega.

Interpretation. These visuals provide suggestive economic interpretation but are not a structural proof of risk premia.

Economic message. The factors span multiple option-relevant states of nature.

6 Short summary

- The paper studies whether large option strategy returns survive an option-specific factor model.

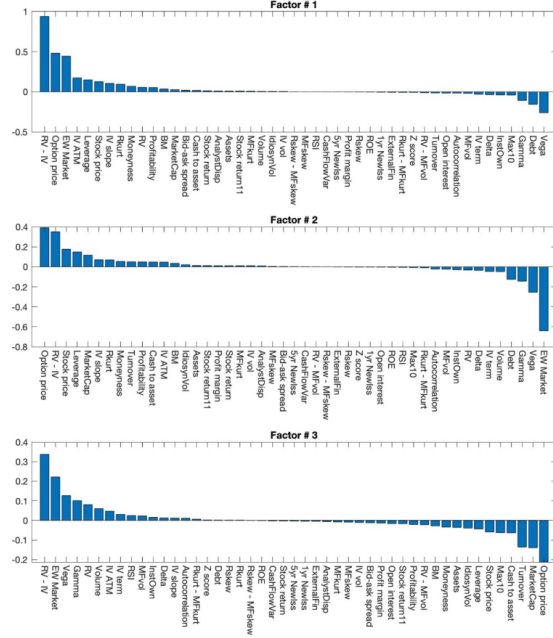


Figure 4
Managed portfolios contribution to IPCA factors
The figure shows average contribution by each managed portfolio to each of the three IPCA factors' means. The sample period is 1996 to 2022.
ALT TEXT: average contribution by each managed portfolio to each of the three IPCA factors' means shows that factor F1 is closely related to RV-IV, factor F2 is inversely linked to the equal weighted market portfolio (EW market), and factor F3 is close to a long-short combination of RV-IV and option price.

Table 7
IPCA factors and economic conditions

	A. Difference of means			B. Linear regression slope		
	F1	F2	F3	F1	F2	F3
Market return	-0.34 (-2.10)	0.35 (2.16)	0.35 (4.29)	-0.45 (-2.96)	0.50 (3.18)	0.22 (4.30)
Δ VIX	-0.59 (-3.95)	0.38 (2.32)	0.40 (4.91)	-0.55 (-5.28)	0.55 (3.64)	0.20 (4.43)
Tail risk	-0.46 (-2.80)	0.26 (1.60)	-0.34 (-2.90)	-0.42 (-4.98)	0.15 (1.61)	-0.25 (-3.88)
Vol risk	1.32 (8.68)	-0.72 (-4.39)	0.27 (-2.60)	0.65 (5.85)	-0.82 (-11.11)	-0.21 (-2.05)
Skew risk	-0.81 (-4.92)	-0.03 (-0.20)	-0.23 (-2.11)	-0.36 (-4.75)	-0.02 (-0.37)	-0.15 (-2.85)
Kurt risk	-0.81 (-4.79)	-0.09 (-0.54)	-0.27 (-2.27)	-0.36 (-5.00)	-0.04 (-0.66)	-0.16 (-3.07)
ICRFac	-0.05 (-0.38)	0.37 (2.65)	0.01 (0.10)	-0.03 (-0.41)	0.28 (3.78)	-0.06 (-0.95)
Average gamma	-0.96 (-6.36)	-0.40 (-2.25)	-0.01 (-0.09)	-0.60 (-7.21)	-0.22 (-2.14)	-0.10 (-1.17)
Average vega	-0.49 (-2.76)	0.23 (1.27)	-0.27 (-2.44)	-0.10 (-1.05)	0.12 (1.13)	-0.20 (-3.28)

The table presents statistics that describe how the 3 IPCA factors are related to economic conditions. In panel A we report average differences in the returns of the factors for months during which economic indicators are higher than the median relative to months when they are below the median. In panel B we report the slope coefficient of univariate regressions of the factor returns on the realization of the same set of indicators. All independent variables are standardized to ensure comparability across variables. The set of economic indicators includes the return of the value-weighted stock market portfolio (Market Return); changes in the VIX index (Δ VIX); level of the tail risk measure of Kelly and Jiang (2014) (Tail risk); the difference between S&P 500 realized volatility, skewness, and kurtosis, and model-free implied volatility, skewness, and kurtosis (Vol risk, Skew risk, and Kurt risk, respectively); the intermediary capital risk factor of He, Kelly, and Manela (2017) (ICRFac); and the cross-sectional equal-weighted average of options gamma and vega measured at the beginning of the option initiation (Average gamma and Average vega). The model-free moments are computed at the beginning of the option initiation while the realized moments are computed over the remaining life of the option. We take negatives of Market return, Vol risk, Kurt risk, and ICRFac to ensure that high values correspond to worse economic conditions. The sample period is 1996 to 2022.

- Forty-six characteristics generate many large gross returns, and 16 remain significant after costs.
- IPCA allows betas to be dynamic functions of characteristics.
- Three latent factors explain most managed-portfolio variation and most strategy returns.
- The average IPCA alpha is about 6 basis points per month.
- RV-IV is the most important characteristic and captures commonality behind many option predictors.
- Restricted-characteristic tests show that IPCA does not merely use each strategy's own signal to price it.

7 Conclusion

7.1 Core conclusion

Large returns to delta-hedged option strategies are not automatically evidence of option-market inefficiency. Once the benchmark allows for latent option factors and dynamic characteristic-based betas, most documented profitability is explained as compensation for systematic option-return exposure.

7.2 Economic mechanism

Option characteristics proxy for covariance with latent option factors. RV-IV is especially important because it captures commonality in many predictors and determines exposure to Factor 1.

7.3 Evidence chain

Table 1 establishes the puzzle; Table 2 validates the IPCA model; Table 3 and Figure 1 show alphas collapse; Figure 2, Table 4, and Table 5 explain the role of characteristics; Figure 3 and Table 6 validate restricted-characteristic pricing; Figure 4 and Table 7 interpret factors.

7.4 Caveats

IPCA is a statistical factor model, not a fully structural economic model. Transaction costs are evaluated after factor estimation. Remaining alphas for RV–IV and IV slope indicate the benchmark is powerful but not complete.

7.5 Final takeaway

Many option predictors earn returns because they proxy for dynamic factor exposures. Under an option-specific IPCA benchmark, most apparent abnormal profits largely disappear.